

The analysis indicates a typical decrease in alloy strength with increasing temperature, attributed to enhanced atomic mobility at higher temperatures. Additionally, the AI agent highlights that the Maresca-Curtin model predictions align with experimental data for some alloys, such as Nb81Mo19, but diverge for others, like Nb95Mo5 and Nb75Mo25. The agent suggests that these discrepancies may result from variations in alloy composition, microstructure, temperature effects, or experimental variability. To address these issues, the agent proposes refining the model to incorporate more detailed microstructural and compositional factors for improved accuracy. Furthermore, the AI agent emphasizes the role of solid solution strengthening, with lattice distortions impeding dislocation motion as a key strengthening mechanism.

This multi-agent collaboration exemplifies the rigorous integration of multiple modalities, combining AI-driven physics predictions with established theoretical frameworks, along with plotting and visualization reasoning. This multifaceted approach not only enhances the precision and depth of the materials design process but also represents a transformative pathway that could revolutionize how materials design is approached and executed.

Plot analysis (GPT-4o)

The analysis of the yield stress versus temperature plots for both binary and ternary Nb-Mo-Ta alloys provides valuable insights into the mechanical behavior of these materials.

Binary Alloys:

- **General Trend**: The yield stress decreases with increasing temperature for all binary alloys, which is consistent with typical metallic behavior.
- **Model Accuracy**: The Maresca-Curtin model predictions align well with experimental data for some alloys, such as Nb81Mo19, but show discrepancies for others, like Nb95Mo5 and Nb75Mo25.
- **Discrepancies**: These may arise from variations in alloy composition, microstructural factors, temperature effects, and experimental variability.
- **Suggestions**: Refining the model to incorporate more detailed microstructural and compositional factors could improve accuracy.

Ternary Alloys:

- **Yield Stress Variation**: The yield stress decreases with temperature for all ternary alloys. Nb20Mo20Ta60 exhibits the highest yield stress, while Nb20Mo60Ta20 shows the lowest.
- **Composition Effects**: High Ta content significantly enhances yield stress due to solid solution strengthening. Balanced compositions provide moderate strengthening, while high Mo content results in lower yield stress.
- **Strengthening Mechanisms**: Solid solution strengthening is the primary mechanism, with lattice distortions impeding dislocation motion. The effectiveness decreases with temperature due to increased atomic mobility.

Overall Insights:

- **Design Implications**: Understanding the balance of alloying elements can aid in designing alloys with desired mechanical properties for specific applications.
- **Temperature Suitability**: These alloys may be more suitable for applications at lower temperatures where higher strength is required.

The results highlight the importance of alloy composition and temperature in determining the mechanical properties of Nb-Mo-Ta alloys, providing a foundation for further research and development in this area.

Figure 8: A summary of the analysis performed by the multi-modal agent of the yield stress versus temperature predictions as shown in Figures 7c and d.

3 Summary and future perspective

First, we developed a GNN model that directly links dislocated crystalline structures, caused by screw dislocations, to material properties in multi-component BCC alloy systems. Despite being trained on a small subset of the massive compositional space in ternary alloys, the model achieves high accuracy in predicting key properties. These include the potential energy change between consecutive screw dislocations and the Peierls barrier, which represents the highest energy at the transition state. One of the main applications of this energy change is the calculation of the solute/screw interaction energy parameter, a fundamental quantity in solute strengthening theories, which our model captures with high precision. The model’s predictions are generated in a matter of seconds, representing a significant improvement over conventional atomistic